Input Variable : x_1 := "Years of Education"
Output : y := "Income"

Relation b/w x_1 and y,

$$y = f(x_1)$$

for some function f.

More Generally if we have n variables,

$$X = (x_1, x_2, \cdots, x_n)$$

{ I/p variables or sometimes called predictors, or independent variables

$$y = f(X) + \varepsilon$$

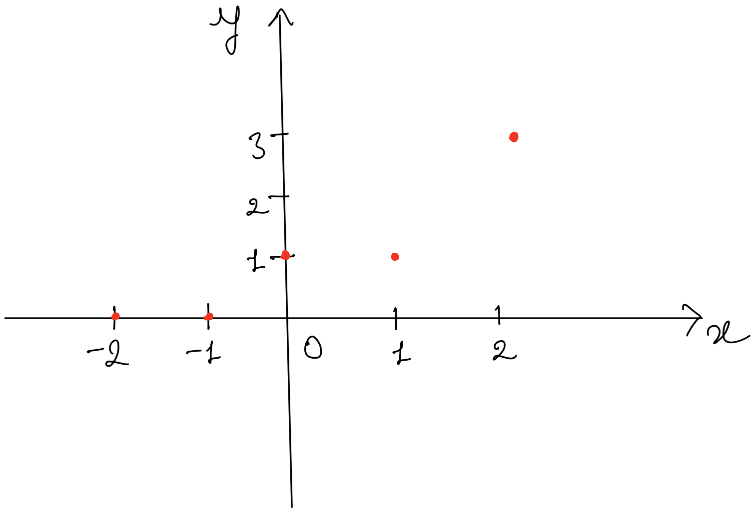
{ y is response variable, or dependent variable.

ε is a random variable such that $E[\varepsilon] = 0$.

The assumption that we start off with is,

$$f(X) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n$$

We observe following data points.



x	y
-2	0
-1	0
0	1
1	1
2	3

Goal: We want to know, that are there some constants β_0 & β_1, such that, we have,

$$y = \beta_0 + \beta_1 x$$

for all possible y & x given above.

This is same as solving the following system of linear equations,

$$\begin{aligned}
\beta_0 - 2\beta_1 &= 0 \\
\beta_0 - 1\beta_1 &= 0 \\
\beta_0 + 0\beta_1 &= 1 \\
\beta_0 + 1\beta_1 &= 1 \\
\beta_0 + 2\beta_1 &= 3
\end{aligned}$$

Simple Linear Model:

We have,

$$Y = \beta_0 + \beta_1 x + \varepsilon$$

where, $E[\varepsilon] = 0$

Thus,

$$E[Y] = \beta_0 + \beta_1 x$$

It is too much to expect to determine β_0 and β_1, thus we tend to estimate them using

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} \quad , \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

where,

$$S_{xy} = \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})$$

and

$$S_{xx} = \sum_{i=1}^{n} (x_i - \bar{x})^2$$

Now as for any other "good" estimator, we expect these to be first and foremost, unbiased, and this is in fact true, i.e.,

$$E[\hat{\beta}_1] = \beta_1$$

and

$$E[\hat{\beta}_0] = \beta_0$$

So far, we have assumed that, $E[\varepsilon] = 0$, now we add one more assumption that $Var(\varepsilon) = \sigma^2$. This implies that the variance of ε is not dependent on the input x_i.

Properties of the Least-Square Estimators; Simple Linear Regression

1. $E[\hat{\beta}_0] = \beta_0$ and $E[\hat{\beta}_1] = \beta_1$.

2. $Var(\hat{\beta}_0) = c_{00}\,\sigma^2$,

   where $c_{00} = \sum x_i^2 / (n S_{xx})$.

3. $Var(\hat{\beta_1}) = c_{11}\sigma^2$,

where $c_{11} = \dfrac{1}{S_{xx}}$.

4. $Cov(\hat{\beta_0}, \hat{\beta_1}) = c_{01}\sigma^2$,

where $c_{01} = \dfrac{-\bar{x}}{S_{xx}}$.

5. An unbiased estimator for σ^2 is

$$S^2 = \frac{SSE}{n-2},$$

where

$$SSE = S_{yy} - \hat{\beta_1} S_{xy}$$

and

$$S_{yy} = \sum (y_i - \bar{y})^2$$

If in addition, the ε_i, for $i = 1, 2, \cdots, n$ are normally distributed,

6. Both $\hat{\beta_0}$ and $\hat{\beta_1}$ are normally distributed.

7. The random variable $\frac{(n-2)S^2}{\sigma^2}$ has a χ^2 distribution with $(n-2)$ df.

8. The statistic S^2 is independent of both $\hat{\beta}_0$ and $\hat{\beta}_1$.

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x	y
29.4	4.25
39.2	5.25
49.0	6.50
58.8	7.85
68.6	8.75
78.4	10.00

1. Fit the model, $Y = \beta_0 + \beta_1 x + \varepsilon$, to the data using the method of least squares.

$S_{xy} = 198.94$

$S_{xx} = 1680.70$

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{198.94}{1680.70} = 0.1184$$

$$\bar{x} = 53.9$$
$$\bar{y} = 7.10$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 0.72$$

2. Find a 95% confidence interval for β_1.

Assuming that ε is normally distributed. Then $\hat{\beta}_0$ and $\hat{\beta}_1$ are also normally distributed.

We want a and b such that,

$$\Pr(a \leq \beta_1 \leq b) = 0.95$$

Now,

$$\frac{\hat{\beta}_1 - \beta_1}{SD(\hat{\beta}_1)}$$

is a standard normal random variable where,

$$SD(\hat{\beta}_1) = \sqrt{Var(\hat{\beta}_1)} = \sqrt{c_{11}\sigma^2}$$

$$= \sigma\sqrt{c_{11}}$$

and $c_{11} = \frac{1}{S_{xx}}$

If we use $S = \sqrt{SSE/(n-2)}$ to estimate σ then in fact,

$$T = \frac{\hat{\beta}_1 - \beta_1}{S\sqrt{C_{11}}}$$

follows a t-distribution with $(n-2)$ df.

Now,

$$Pr(a \leq T \leq b) = 0.95$$

$$a = qt(0.025, 4) = -2.77$$
$$b = -a = 2.77$$

Since,

$$SSE = S_{yy} - \hat{\beta}_1 S_{xy}$$
$$= 23.60 - 0.1184 \times 198.94$$
$$= 0.046$$

$$S = \sqrt{\frac{SSE}{4}} = 0.176$$

Therefore,

$$Pr\left(-2.77 \leq \frac{0.1184 - \beta_1}{0.176\sqrt{0.0006}} \leq 2.77\right) = 0.95$$

$\therefore$ $\beta_1 \in 0.1184 \pm 2.77 \times 0.176\sqrt{0.0006}$ with 95% probabability.

$$[0.11, \quad 0.43]$$